WORKSHEET

Mixed differentiation problems

1 Find the gradient function of the following.

a $f(x) = x^4 - 6x^3 - 2x^2 + 5x + 9$

b $y = 3(2x - 5)^3$

c $g(x) = (12 - 5x^2)^6$

d $h(k) = \dfrac{x^2 - 3}{5x + 8}$

e $p(q) = 3q^2(q^3 - 5q^2 + 2q - 1)$

f $x(t) = 4t^2(t^2 - 10)^4$

2 Differentiate the following functions.

a $3x^4 - 5x^3 + \frac{1}{2}x^2 + 4x - 7$

b $(2x + 3)^5$

c $\dfrac{4x - 1}{5x + 3}$

d $8x^5 + 2x^3 - 5x^2 + 3$

e $(7x^4 + 3x)(x^5 - 4x^2 + 9)$

f $4(3 - 5x)^3$

g $3 + 4x^3 - 7x^5 - x^8$

h $(4x^2 - 2x + 1)(5x - 7)$

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i $\dfrac{2x^3 - 3}{x^2 - 5x + 1}$

j $\dfrac{1}{(5x + 7)^4}$

k $(7x^2 + 2x - 1)(x^2 - 5x + 3)$

l $7x^{10} + 8x^7 - 2x^6 - 3x^4 - 4$

m $\dfrac{4}{(10 - x)^7}$

n $\dfrac{3x^2 + x - 1}{2x + 5}$

o $(x^2 - 1)(3x^3 - 5x^2 + x - 2)$

p $(2x - 3)^2(5x + 2)^5$

q $x^6 - 4x^3 + 7x$

r $\frac{1}{2}(4x + 1)^8 + 3x - 2$

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Answers

1 a $f'(x) = 4x^3 - 18x^2 - 4x + 5$

b $\frac{dy}{dx} = 18(2x - 5)^2$

c $g'(x) = -60x(12 - 5x^2)^5$

d $h'(k) = \frac{5k^2 + 16k + 15}{(5k + 8)^2}$

e $p'(q) = 6q(q^3 - 5q^2 + 2q - 1) + 3q^2(3q^2 - 10q + 2)$

f $x'(t) = 8t(t^2 - 10)^4 + 32t^3(t^2 - 10)^3$

2 a $12x^3 - 15x^2 + x + 4$

b $10(2x + 3)^4$

c $\frac{17}{(5x + 3)^2}$

d $40x^4 + 6x^2 - 10x$

e $(28x^3 + 3)(x^5 - 4x^2 + 9) + (7x^4 + 3x)(5x^4 - 8x)$

f $-60(3 - 5x)^2$

g $12x^2 - 35x^4 - 8x^7$

h $(8x - 2)(5x - 7) + 5(4x^2 - 2x + 1)$

i $\frac{2x^4 - 20x^3 + 6x^2 + 6x - 15}{(x^2 - 5x + 1)^2}$

j $\frac{-20}{(5x + 7)^5}$

k $(14x + 2)(x^2 - 5x + 3) + (7x^2 + 2x - 1)(2x - 5)$

l $70x^9 + 56x^6 - 12x^5 - 12x^3$

m $\frac{28}{(10 - x)^8}$

n $\frac{6x^2 + 30x + 7}{(2x + 5)^2}$

o $2x(3x^3 - 5x^2 + x - 2) + (x^2 - 1)(9x^2 - 10x + 1)$

p $4(2x - 3)(5x + 2)^5 + 25(2x - 3)^2(5x + 2)^4$

q $6x^5 - 12x^2 + 7$

r $16(4x + 1)^7 + 3$

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